

Amazon Route 53 - Service Overview

Introduction

Amazon Route 53 is a highly available and scalable cloud Domain Name System (DNS) web service. Route 53 is designed to give developers and businesses an extremely reliable and cost-effective way to route end users to internet applications, and takes its name from port 53, the standard port for DNS traffic.

Hosted Zones

A hosted zone is a container for records that define how traffic is routed for a domain (e.g. example.com) and its subdomains. Route 53 supports both public hosted zones (for routing internet traffic) and private hosted zones (for routing traffic within one or more VPCs, without exposing DNS records to the public internet).

Record Types

Route 53 supports standard DNS record types, including:

- A record: maps a domain name to an IPv4 address.
- AAAA record: maps a domain name to an IPv6 address.
- CNAME record: maps a domain name to another domain name.
- Alias record: an AWS-specific extension that behaves like a CNAME but can be used at the zone apex (root domain), and can point directly at AWS resources like an ALB or CloudFront distribution, at no additional query cost.
- MX record: specifies mail servers responsible for accepting email for the domain.
- TXT record: holds arbitrary text data, commonly used for domain verification or SPF/DKIM email authentication records.

Routing Policies

Route 53 supports multiple routing policies beyond simple one-to-one mapping:

- Simple routing: a single resource performs a given function for the domain.
- Weighted routing: routes a specified percentage of traffic to multiple resources, useful for A/B testing or gradual rollouts.
- Latency-based routing: routes traffic to the region providing the best latency for the end user.
- Failover routing: routes traffic to a primary resource, automatically failing over to a secondary resource if the primary becomes unhealthy, based on Route 53 health checks.
- Geolocation routing: routes traffic based on the geographic location of the requester.

Health Checks

Route 53 can perform health checks against endpoints (an IP address or domain name), monitoring for HTTP/HTTPS/TCP status, and can trigger DNS failover when an endpoint becomes unhealthy, or feed CloudWatch alarms for broader operational visibility.

Domain Registration

In addition to DNS hosting, Route 53 can also act as a domain registrar, allowing you to purchase and manage domain names directly within AWS, integrating registration, DNS management, and routing into a single console.

TTL (Time To Live)

Each DNS record has a TTL value, specifying how long resolvers are permitted to cache the record before querying Route 53 again. Lower TTLs allow faster propagation of DNS changes (useful before a planned failover or migration), at the cost of more frequent queries and slightly higher query charges.

How DNS Resolution Reaches Route 53

When a user's browser requests a domain, their local resolver (often provided by their ISP or a public resolver like 8.8.8.8) queries the DNS root servers, which direct it to the Top-Level Domain (TLD) servers (e.g. for .com), which in turn direct it to the authoritative name servers for that specific domain. When a domain's hosted zone is created in Route 53, AWS assigns it four authoritative name servers, and the domain's registrar must be configured to delegate to those name servers for Route 53 to actually control resolution for the domain.

Pricing Model

Route 53 charges a small monthly fee per hosted zone (roughly \$0.50/month for the first 25 zones), plus a per-million-queries charge that varies by routing policy type (simple queries are cheaper than latency-based or geolocation queries), plus separate charges for health checks and domain registration/renewal if using Route 53 as the registrar.

Common Use Cases

- Routing a custom domain to an Application Load Balancer or CloudFront distribution via an Alias record
- Implementing automated failover between a primary and disaster-recovery region
- Private DNS resolution for internal microservice-to-microservice communication within a VPC

Best Practices

- Use Alias records rather than CNAME records when pointing at AWS resources, since Alias records work at the zone apex and incur no additional query charges.
- Lower TTLs temporarily before a planned cutover or migration, then raise them again afterward to reduce ongoing query volume.
- Pair failover routing policies with Route 53 health checks to achieve automated, DNS-level disaster recovery.